

Parth N. Kheni

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EDUCATION

Boston University

Boston, MA

B.S. Computer Engineering, Concentration in Machine Learning

Expected Dec 2026

- Related Coursework: Differential Equations, Multivariable Calculus, Microeconomics, Macroeconomics, Statistics, Linear Algebra, Electric Circuits, Discrete Math, Data Structures

Rutgers University

New Brunswick, NJ

B.S. Computer Engineering | GPA: 3.97 | Dean's List

Sept 2023 – May 2024

SKILLS

Programming: Python, SQL, MATLAB, C++ (primary), Java, C#, Objective-C, OOP, Kotlin

Libraries/Frameworks: NumPy, Matplotlib, Pandas, scikit-learn, TensorFlow, Retrofit/Gson, Android Jetpack

Certifications: Electronic Arts Software Engineering Job Simulation and Python for Everybody Specialization

Cloud/Tools: Azure AD, GitHub, Microsoft 365, Arduino, Android Studio, API, Power BI, WorkManager

Machine Learning: Predictive Modeling, Statistical Modeling, Time-Series Analysis, Anomaly Detection

EXPERIENCE

Beth Israel Deaconess Medical Center (BIDMC), Machine Learning Intern

Sept 2025 – Present

- Build a GPU-enabled surgical robotics simulation workflow for the da Vinci Research Kit (dVRK, a research platform based on the da Vinci surgical robot) using ROS and Gazebo/AMBF on BU's Shared Computing Cluster (SCC) by containerizing dependencies with Singularity/Apptainer and creating reliable scripts to launch and view simulations remotely.
- Prototype incision-closure tasks in NVIDIA Isaac Sim/ORBIT-Surgical by setting up a reproducible Omniverse/Isaac environment (fixed versions and dependencies) and organizing the required simulation assets for surgical manipulation.
- Improve ML autonomy training reliability by diagnosing a preprocessing mismatch that reduced performance from ~90% to ~30%, curating/labeling demonstration data (e.g. ~80 JIGSAWS trajectories), and collaborating with Johns Hopkins to acquire new dVRK kinematic demos to standardize multimodal inputs (vision + kinematics).

Boston University Center for Space Physics, Machine Learning Research Assistant

Apr 2025 – Present

- Build end-to-end Python pipelines for POES and OMNI time-series data, generating model-ready datasets with 7M+ 2-second satellite samples per month by ingesting NetCDF files, interpolating gaps, enforcing L-shell (3–9) and MLT quality cuts, and joining with 4-hour windows of 1-minute solar-wind drivers.
- Improve accuracy and stability of energetic electron precipitation (E4_0 and E4_90) forecasts, reducing test MAE/RMSE and tightening correlation vs earlier baselines by training Keras neural networks and Ridge regressors with standardized features and time-ordered train/val/test splits.
- Increase interpretability of space-weather predictions across different storm conditions and locations in the radiation belts, improving generalization during rare, high-flux events, by testing different time resolutions (2-second vs 1-minute data), quantile-based flux aggregations, and simple features that capture local time around Earth.

Blue Leaf Technologies, IT Consultant

Jul 2023 – Aug 2023

- Improved employee sign-in and credential flow (25% faster logins, 40% fewer credential bottlenecks, 50% less manual tracking) by deploying Azure AD + Windows Hello for Business, implementing Cloud Kerberos Trust with domain controllers, and automating monitoring/data collection with Python.

PROJECTS

Whack-a-Mole Reaction Game on FPGA (Nexys4 DDR, Verilog), FPGA Design Engineer

Sept 2025 – Present

- Design a real-time Whack-a-Mole reaction game on a Nexys4 DDR FPGA, enabling 30-second multi-difficulty rounds with 5 LED “moles” and per-mole switch/hammer input by implementing debounced button I/O, difficulty timing, random mole selection, and top-level integration in Verilog.
- Improve reliability and integration, delivering a glitch-free demo by owning input and game-control modules, coordinating interfaces across a 4-person team, and iterating with Vivado simulation, timing analysis, and on-board testing.

Personal Finance App, Android Developer

Jan 2025 – May 2025

- Built in Android Studio with Java, Retrofit/Gson, and the Finnhub.io API to fetch and display real-time stock symbols and quotes in RecyclerView and MPAndroidChart line charts.
- Engineered portfolio persistence using SharedPreferences, enabled navigation from MainActivity to BrowseStocksActivity for seamless stock selection, and integrated WorkManager background tasks with Android Notifications for price alerts.

ECG Data Analysis, Data Science Project

Sept 2024 – Dec 2024

- Built an ECG arrhythmia detection pipeline (>90% accuracy) by cleaning raw signals in Python/Pandas, applying digital filtering and time/frequency feature engineering, and training/validating SVM and Random Forest models in scikit-learn with clear result visualizations (Matplotlib/Seaborn).

ACTIVITIES

BU Baja Society of Automotive Engineers, Head Engineer of Computer Systems

Sept 2024 – Present

- Lead a team of three engineers to implement GPS navigation, real-time communications, and telemetry systems for an off-road racing car using Agile/DevOps practices, boosting accuracy and performance.

BU Jalwa Dance Troupe, Secretary

Sep 2024 – Present